MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 2AP1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	Vdc	ft
Maximum:	6.9	0	660	550	1,100	---	1.5	1.0	-125	10,000
Minimum:	5.7	-125	---	---	500	2.0	---	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	1,000	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Qualification inspection						
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	1.63	Inches
(A) 5216	Cathode illumination	-		---	---	---	---
5248	Deflection factor uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Control grid to all	Cg1	---	11.0	pF
			Cathode to all	Ck	---	7.5	pF
			D1 to D2	C1D2	---	2.0	pF
			D3 to D4	C3D4	---	2.0	pF
			D1 to all	CD1	---	12.5	pF
			D3 to all	CD3	---	13.0	pF
			D1 to all, except D2	CD1	---	12.0	pF
			D2 to all, except D1	CD2	---	12.0	pF
			D3 to all, except D4	CD3	---	10.0	pF
			D4 to all, except D3	CD4	---	9.0	pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
(A) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1	Light = 2 ftL	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5221	Light output	1		Light	2.0	---	ftL
5223	Modulation	-	Light = 2 ftL	ΔEc	---	50	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	10	mm
5231	Spot displacement (leakage)	-		Displ	---	5	mm
5244	Grid cutoff voltage	-		Ec1	-30	-90	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-50	10	μAdc
(A) 5201	Electrode current (cathode)	-	Light = 2 ftL	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	-	+ 3D4, pin No. 1	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5101	Angle between traces	-		Angle	86	94	Degrees
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-		---	---	---	---
5226	Line width "A" (electrostatic deflection)	2	Light = 2 ftL	Width	---	0.45	mm
5226	Line width "B" (electrostatic deflection)	3	Light = 2 ftL	Width	---	0.55	mm
5246	Focusing voltage at cutoff	-		Eb1	200	300	Vdc
(A) 5246	Focusing voltage (zero-bias)	-		Eb1	140	300	Vdc
5248	Deflection factor	-	1D2	DF	195	265	Vdc/in.
5248	Deflection factor	-	3D4	DF	167	225	Vdc/in.
5251	Heater-cathode leakage current	-		---	---	---	---
(A) 1101	Secureness of base, cap, or insert			---	---	---	---
(A) 1111	Base pin solder depth	-		---	---	---	---
(A) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group B; Light = 2 ftL; Eb2 = 1,100 Vdc; t = 500 hours	---	---	---	---
---	Life-test end points:						
5226	Line width "A"	-	Light = 2 ftL	Width	---	0.50	mm
5226	Line width "B"	-	Light = 2 ftL	Width	---	0.65	mm
5223	Modulation	-	Light = 2 ftL	ΔEc	---	50.0	Vdc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. The deflecting plates shall be returned to anode No. 2 through a minimum of 2.5 megohm resistors. The light output shall be set at 2 ftL. The high frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction of high frequency scanning. Readjustment shall be made for best overall focus. The tubes shall be observed for deflection defocusing, astigmatism, and spot ellipticity observable to the eye, as evidenced by fuzziness due to lack of sharpness of trace (usually around the edges), bow-tieing (irregular widths of any line when observed at different points) and bowing of trace other than that normally caused by curvature of the bulb. This test for focus is to be made in addition to the line width measurement.
3. The same conditions shall be set up as described in note 2 except that the connection of deflecting elements to the low and high frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 2 conditions. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 2.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

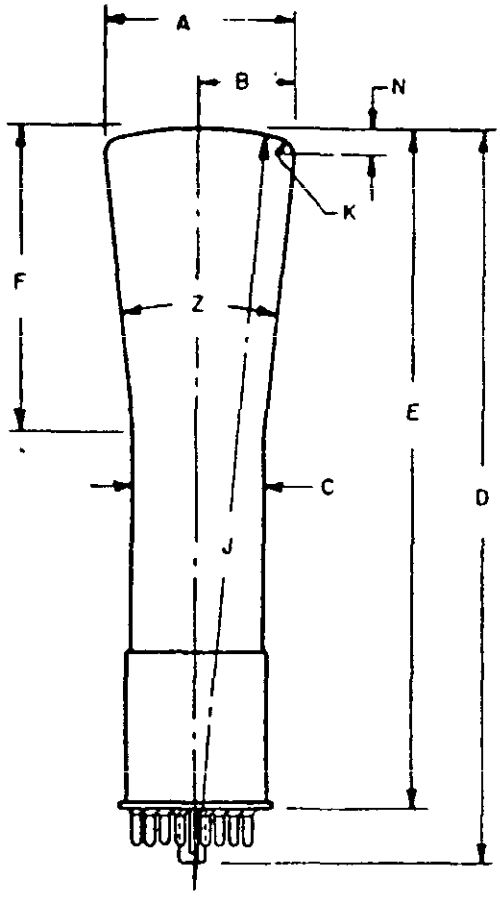
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

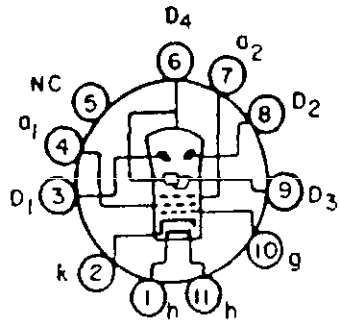
Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	1.937 (49.20)	2.063 (52.40)
B	Note a	
C	1.312 (33.32)	1.438 (36.53)
D	7.250 (184.15)	7.626 (193.70)
Reference dimensions (see note d)		
E	6.875 (174.63)	
F	3.062 (77.77)	
J	8.000 (203.20) R	
K	.188 (4.78) R	
N	.225 (5.72)	
Z	12°37'	

NOTES:
a. The minimum useful screen radius shall not be less than .875 inches (22.23 mm).
b. The base shall be a large wafer magnal 11-pin, with flared sleeve, (B11-36) type or a small magnal 11-pin (B11-33) type.
c. The bulb shall be a J16A type.
d. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing for electron tube type 2AP1A.

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